Supplementary Material F: Preliminary Diagnostic Phase

Baseline AI-Literacy Profile of the Institutional Context ($N = 121$)

Note. This document reports the preliminary diagnostic referenced in the Introduction of the main manuscript. It was compiled from the diagnostic phase of the research program within which the present study was designed. Anonymized for peer review.

F1. Purpose and Method

Before designing the intervention, a diagnostic survey was conducted to characterize the baseline AI-literacy profile of the institutional context: a multidisciplinary public university in Lambayeque, Peru. In late 2024, a total of 121 undergraduate students from more than 20 academic programs completed a questionnaire covering the four AI-literacy dimensions subsequently operationalized in the main study: conceptual understanding, application of AI techniques (technical skills), ethical and social implications, and collaboration and knowledge transfer. The questionnaire was content-validated through expert judgment using Aiken's V (the validation process that produced the initial version of the study instrument; see Section 2.3.1 of the manuscript). Responses were categorized into four performance levels (Deficient, Fair, Good, Very good).

F2. Results

Table F1: Distribution of performance levels by AI-literacy dimension ($N = 121$; frequencies with percentages).

Dimension	**Deficient**	**Fair**	**Good**	**Very good**
Understanding of basic AI concepts	30 (24.8%)	63 (52.1%)	28 (23.1%)	0 (0.0%)
Application of AI techniques	11 (9.1%)	52 (43.0%)	55 (45.5%)	3 (2.5%)
Analysis of ethical and social implications of AI	49 (40.5%)	29 (24.0%)	42 (34.7%)	1 (0.8%)
Collaboration and knowledge transfer	42 (34.7%)	55 (45.5%)	23 (19.0%)	1 (0.8%)

Note. Row totals equal 121 in all cases. Dimension labels are English renderings of the original diagnostic instrument; they correspond to dimensions D1–D4 of the main study.

F3. Implications for the Intervention Design

The diagnostic profile is the empirical footprint of the instrumental–integral gap that motivates the main study. Students performed comparatively well in the applied, tool-oriented dimension — only 9.1% were classified as Deficient in the application of AI techniques, and 48.0% reached Good or Very good — while the integral dimensions lagged markedly: 40.5% were Deficient in the analysis of ethical and social implications and 34.7% in collaboration and knowledge transfer, with conceptual understanding in an intermediate position (24.8% Deficient; no student reached Very good). In other words, instrumental competence was already developing through everyday tool use, whereas ethical reasoning, collaborative capacity, and conceptual depth were not developing incidentally.

This profile informed the design of the intervention in three ways: (a) it justified addressing all four dimensions simultaneously rather than focusing on technical instruction; (b) it motivated the deliberate strengthening of the ethical and collaborative components of the instrument and the program (see Sections 2.3.1 and 2.4 of the manuscript); and (c) it grounded the room-for-growth component of hypothesis H1b, since baseline deficits were largest in precisely the ethical and collaborative dimensions.